Question bank

Semiconductor and Quantum Physics 26SOST101

1. **Unit 3**
    - Definition: wave particle duality, wave function and its significance, tunneling effect, quantum computing
    - Derivation: Time independent Schrodinger Equation.
    - Derivation: Particle in 1D box

2. **Unit 4**
    - Definition: Spontaneous and Stimulated absorption/emission, metastable state, population inversion, He-Ne Laser, Ruby Laser
    - Optical Fiber, Principle and classifications
    - Derivation: Numerical Aperture and Acceptance angle

3. **Tutorial sheets discussed in class**